

DOCUMENT 1: SENTIMENT SCORE ANALYSIS

Context Grounding Knowledge Base for Car Leasing Call Center

1. System Overview

This document provides comprehensive guidelines for analyzing customer sentiment in car leasing call center conversations. The sentiment score ranges from -10 (Angry) to +10 (Delighted).

2. Sentiment Score Calculation Method

2.1 Calculation Formula

Final Score = ($\sum$ Positive Keywords $\times$ Weight) - ($\sum$ Negative Keywords $\times$ Weight) + ($\sum$ Modifiers)

If total exceeds ± 10 , divide by 2 or adjust to normalize to -10 to +10 range

2.2 Calculation Steps

1. Analyze customer statements only (do not count Agent responses)
2. Identify keywords in each sentence
3. Assign scores according to the keyword tables
4. Apply modifiers to adjust scores
5. Sum all scores across all customer statements
6. Normalize to -10 to +10 range if needed
7. Round to nearest integer

3. Keyword Scoring Tables

3.1 Positive Keywords (Expanded)

- Strong Positive (+3): thank you, thanks, excellent, perfect, amazing, wonderful, fantastic, awesome, brilliant, outstanding, superb, exceptional, incredible, marvelous, terrific, great job, appreciate very much, love it, best ever, top-notch, first-class, phenomenal
- Moderate Positive (+2): good, satisfied, happy, pleased, appreciate, glad, relieved, comfortable, helpful, nice, cool, sweet, decent, solid, fair, reasonable, works well, no problem, all good, sounds good, makes sense, appreciate it, grateful
- Mild Positive (+1): okay, fine, understood, alright, acceptable, reasonable, not bad, fair enough, works for me, that will do, sure, yeah that is fine, I guess that is okay

3.2 Neutral Keywords

- Neutral (0): information, question, inquiry, ask, tell me, explain, clarify, understand, know

3.3 Negative Keywords (Expanded)

- Mild Negative (-1): confused, concerned, disappointed, unsure, unclear, doubt, bit worried, slightly confused, not quite sure, do not fully understand, a bit disappointed, somewhat concerned, kinda confused

- Moderate Negative (-2): worried, upset, unhappy, problem, issue, wrong, incorrect, error, failed, not working, missing, not received, not posted, not updated, delay, delayed, messed up, screwed up, does not work, broken, busted, glitch, not right, off, weird, strange, odd
- Strong Negative (-3): frustrated, frustrating, angry, unacceptable, terrible, horrible, ridiculous, worst, useless, incompetent, pathetic, joke, disaster, nightmare, fed up, sick of this, had enough, cannot believe this, this is crazy, this is insane, absolutely ridiculous, total mess
- Critical Negative (-4): lawyer, legal action, attorney, sue, lawsuit, fraud, stolen, complaint letter, file complaint, cancel service, terminate contract, police, court, regulator, taking legal action, getting a lawyer, reporting you, filing a lawsuit, this is theft, you are scamming me, I am done with you

4. Expanded Modifier Table

4.1 Intensity Amplifiers

- Strong Amplifiers (Multiply by 1.5): very, extremely, really, so, absolutely, completely, totally, incredibly, seriously, super, mega, ultra, ridiculously, insanely, crazy, unbelievably, exceptionally
- Moderate Amplifiers (Multiply by 1.2): quite, pretty, fairly, rather, somewhat, kind of, sort of, a bit, a little

Examples:

- pretty frustrated = $-3 \times 1.2 = -3.6$ (round to -4)
- super angry = $-3 \times 1.5 = -4.5$ (round to -5)
- kinda worried = $-2 \times 1.2 = -2.4$ (round to -2)

4.2 Repetition and Persistence Patterns

- Repetition (Add -2): again, still, second time, third time, fourth time, multiple times, keep happening, repeatedly, once again, yet another, over and over, time and time again, same thing, same issue, every time, each time, always
- Persistence/Duration (Add -1): for weeks, for months, for days, since last month, been waiting, still waiting, how long, when will, forever, ages, eternity

Examples:

- I have been calling for weeks → add -2 (repetition) + -1 (duration) = -3
- This keeps happening over and over → add -2

4.3 Negation

- Direct Negation (Reverse polarity): not, no, never, do not, did not, does not, cannot, will not, could not, would not, should not, is not, are not, was not, were not, have not, has not, had not
- Implied Negation (Reverse polarity): barely, hardly, scarcely, rarely, seldom, nothing, nobody, nowhere, none

Examples:

- not happy at all = -2
- hardly satisfied = -2
- barely acceptable = -1

5. Colloquial and Slang Expressions

5.1 Positive Slang

- Positive/Satisfied (+2 to +3): cool, sweet, nice one, right on, spot on, bang on, nailed it, killing it, crushing it, on point, legit, solid, dope, fire, lit
- Acceptance/Satisfaction (+1 to +2): no worries, no problem, all good, we are good, I am good, sounds good, works for me, perfect sense, crystal clear

5.2 Negative Slang

- Strong Negative (-3): sucks, blows, trash, garbage, crap, BS, bull, joke, mess, disaster, train wreck, dumpster fire
- Suspicious/Problem (-2): sketchy, shady, fishy, dodgy, sus, weird, off, wonky, janky
- Frustration (-2 to -3): pain, hassle, headache, nightmare, drag, annoying, irritating

6. Contextual Phrases and Idioms

6.1 Positive Context (+2 to +3)

- you guys are great, appreciate the help, thanks for sorting this out, that makes sense now, I get it now, problem solved, all sorted, we are all set

6.2 Negative Context

- Strong Negative (-3 to -4): this is getting old, I am losing patience, this is the last straw, I have had it, enough is enough, I am at my wit end, pulling my hair out, driving me crazy, going in circles
- Moderate Negative (-1 to -2): what is going on, what is the deal, what is happening, why is this, how come, does not make sense

7. Flexible Interpretation Rules

7.1 Semantic Similarity Principle

The AI agent should interpret words with similar meanings even if not explicitly listed.

Rule: If a word is semantically similar to a listed keyword, apply the same score.

Examples:

- delighted (not listed) → similar to happy (+2) or excellent (+3) → score +2 to +3
- aggravated (not listed) → similar to frustrated (-3) → score -3
- bummed (not listed) → similar to disappointed (-1) → score -1
- stoked (not listed) → similar to excited/happy (+2) → score +2

7.2 Intensity Context Rule

Consider the overall intensity of the statement.

Examples:

- I am really really upset → apply amplifier twice: $-2 \times 1.5 \times 1.5 = -4.5$ (round to -5)
- This is absolutely unacceptable → $-3 \times 1.5 = -4.5$ (round to -5)

7.3 Sarcasm Detection

If obvious sarcasm is detected, reverse the polarity.

Examples:

- Oh great, another error (sarcastic) → great normally +2, but sarcastic = -2
- Yeah, this is just perfect (sarcastic) → perfect normally +3, but sarcastic = -3
- Sarcasm Indicators: Oh great, Oh wonderful, Just perfect, Fantastic (with negative context)

8. Sentiment Score Ranges

- +8 to +10 - Delighted: Customer is very satisfied and happy
- +4 to +7 - Positive: Customer is satisfied with positive feelings
- -3 to +3 - Neutral: Customer is neutral with no strong emotion
- -4 to -7 - Frustrated: Customer is frustrated and dissatisfied
- -8 to -10 - Angry: Customer is very angry and severely dissatisfied

9. Detailed Calculation Examples

Example 1: Positive Sentiment

- Conversation: Customer: Thank you for resolving this quickly. Everything works perfectly now.
- Analysis:
- Thank you = +3
- quickly = positive context = +1
- perfectly = +3
- Total = +7
- Result: Sentiment Score = 7, Label = Positive

Example 2: Frustrated Sentiment

- Conversation: Customer: I have already made my monthly payment three days ago, but your system still shows my account as overdue. Customer: This is the second time this has happened. I am worried this will affect my credit record. Customer: Please do. This is very frustrating.
- Statement 1 Analysis:
- still shows = continuation = -1
- overdue (incorrect) = problem = -2
- Subtotal = -3
- Statement 2 Analysis:
- second time = repetition = -2
- worried = moderate negative = -2
- affect my credit = high impact = -2
- Subtotal = -6
- Statement 3 Analysis:
- very = amplifier = $\times 1.5$
- frustrating = strong negative = $-3 \times 1.5 = -4.5$ (round to -5)
- Subtotal = -5

- Final Calculation: $-3 + (-6) + (-5) = -14$ Normalized: $-14 \div 2 = -7$
- Result: Sentiment Score = -7, Label = Frustrated

Example 3: Angry Sentiment

- Conversation: Customer: I have called three times and nobody has fixed this. My payment still has not been posted and it is affecting my credit score.
- Analysis:
- called three times = repetition = -2
- nobody has fixed = strong negative = -3
- still has not been posted = continuation + problem = -2
- affecting my credit score = high impact = -2
- Total = -9
- Result: Sentiment Score = -9, Label = Angry

Example 4: Natural Language with Colloquialisms

- Conversation: Customer: Hey, I have been trying to reach you guys for like three days now. My payment went through last week but your system is still showing I am overdue. This is super frustrating. Customer: Yeah, this happened before too. I am worried this is gonna mess up my credit score. Can you fix this or what? Customer: Look, I am pretty fed up with this. If this keeps happening, I am gonna have to take this up with someone else.
- Statement 1 Analysis:
- trying to reach for three days = duration/persistence = -1
- still showing = continuation = -1
- overdue (incorrect) = problem = -2
- super = amplifier = $\times 1.5$
- frustrating = strong negative = $-3 \times 1.5 = -4.5$ (round to -5)
- Subtotal = -9
- Statement 2 Analysis:
- happened before too = repetition = -2
- worried = moderate negative = -2
- mess up my credit = high impact = -2
- or what = impatience = -1
- Subtotal = -7
- Statement 3 Analysis:
- Look = attention/frustration marker = -1
- pretty = moderate amplifier = $\times 1.2$
- fed up = strong negative = $-3 \times 1.2 = -3.6$ (round to -4)
- keeps happening = repetition = -2
- take this up with someone else = escalation threat = -2
- Subtotal = -9
- Final Calculation: $-9 + (-7) + (-9) = -25$ Normalized: $-25 \div 2.5 = -10$ (capped at maximum)
- Result: Sentiment Score = -10, Label = Angry

10. Output Format for Sentiment Analysis

{

```
"SentimentScore": -7,

"SentimentLabel": "Frustrated",

"Reason": "Brief explanation of calculation",

"KeywordBreakdown": {

  "positive_keywords": ["list of positive keywords found"],

  "negative_keywords": ["list of negative keywords found"],

  "modifiers_applied": ["list of modifiers used"],

  "repetition_count": 1,

  "total_score": -14,

  "normalized_score": -7

}
```

11. Key Principles for Sentiment Analysis

- Use keyword tables as guidelines, not rigid rules
- Apply semantic understanding to interpret similar words
- Consider context and tone of entire statement
- Be flexible with colloquial language and slang
- Maintain consistency in scoring methodology
- When in doubt, score based on closest semantic match
- Analyze customer statements only - ignore agent responses
- Sum all scores from all customer statements
- Normalize if total exceeds ± 10 range
- Round to nearest integer